

43/50

Answer sheet

Full name: _____

Part I:

Section _____

Part II:

Part III:

Part IV:

1. a) Reduce Energy Efficiency
b) Using Solar Energy
c) Efficient means is through implementation (choosing them to decrease power)
d) Increase the use of Renewable Energy
2. a) Most of them are Trans-boundary
b) Most of them start from highlands of the mountain
c) Most of them are not suitable for transportation
d) Water source is not suitable for drinking without treatment
3. a) Agricultural Impact
b) Transportation and communication impact
c) Impact on Culture
d) Impact on Settlement
4. a) Moderate or high rainfall (Seasonal)
b) Low mean annual T° (temperature) (< 20°C)
c) Free from tropical diseases
5. a) Building flood defenses
b) Restoration
c) Creating water resources

Part V: Sea level T° 45°C

$$X = 1000 = \frac{1000}{1000} = 1$$

$$dH = 1 \times 100 = 100 \quad 45^\circ \text{C} = 30^\circ \text{C}$$

$$wet = 1 \times 5 = 5 \quad 45^\circ \text{C} = 5^\circ \text{C}$$

$$Y = 2000 = \frac{2000}{1000} = 2$$

$$dH = 2 \times 100 = 200 \quad 45^\circ \text{C} = 25^\circ \text{C}$$

$$wet = 2 \times 5 = 10 \quad 45^\circ \text{C} = 10^\circ \text{C}$$

$$Z = 4000 = \frac{4000}{1000} = 4$$

$$dH = 4 \times 100 = 400 \quad 45^\circ \text{C} = 15^\circ \text{C}$$

$$wet = 4 \times 5 = 20 \quad 45^\circ \text{C} = 20^\circ \text{C}$$

$$45^\circ \text{C} = 22.5^\circ \text{C}$$

D. 24 hours of darkness at South Pole, 24 hours day light at North Pole; 12 hours day light at Equator

19. In which season do the northeasterly winds dominantly prevail to the Ethiopian landmass?

- A. Summer B. Winter C. Autumn D. Spring

20. Which one of the following is the major source of greenhouse gas in the developed country?

- A. Emission of different gases from industries C. Land resources degradation
B. High population growth rate D. None of the above

21. Which one of the following is not a source of greenhouse gases?

- A. Wastes in landfills. C. Ruminant digestion and manure management
B. Afforestation D. Clearing of land for agriculture

22. Currently the most threatened lake due to water extraction for soda ash production is

- A. Abiyata B. Shala C. Langano D. Abaya

23. Of the following areas which one is appropriate site to find varieties of migratory birds?

- A. Lake Abjiata C. Savanna land
B. Semien mountain D. Lake Koga (koka)

24. Which of the following is a non-moisture-laden wind?

- A. Southeasterlies C. Northeast trade wind
B. Equatorial westerlies D. Guinea monsoon

PART IV: Short Answer Questions. (0.5 mark each sub item)

1. List down four climate change mitigation measures

- a) Reduce CO₂ emissions
b) Use more solar energy
c) Improve energy efficiency and use renewable energy
d) Plant more trees

2. Mention at least four major characteristics of the Ethiopian Rivers;

- a) Most of them start from highlands
b) Most of them are short - lived
c) Most of them are seasonal
d) Most of them are polluted

3. Indicate the impacts of Relief on Biophysical and Socioeconomic Conditions of a place;

- a) Agriculture
b) Transportation and communication
c) Settlements

~~Handwritten notes~~ ^{in pencil on}

List down three major characteristics of highland areas of Ethiopia:

a) Magnitude of kinetic energy with first 3000 rpm.

b) low pressure region $70 \text{ } ^\circ\text{C}$

c) type from barren degenerate

5. List down three climate change adaptation measures

a) $\text{butyramide} + \text{stronger base} \rightarrow \text{butyrate} + \text{H}_2\text{O}$

b) Preferred name

c)

Part V: Calculations: Show the necessary steps briefly and neatly (6 mark)

1. Suppose the temperature at sea level is 45°C and there are three places, X, Y, and Z at three altitude (Elevations), 1000 meter below sea level, 2000 meter above sea level, and 4500 meter above sea level respectively. Calculate the temperature for place X, Y, and Z for Environmental, Wet and Dry adiabatic lap's rate.

10. Which of the following is a **wrong** combination?
- A. Tigray Plateau-Mount Guna C. North Central Massifs-Mount Tsibet
 B. Shewa Plateau- Mount Guraghe D. The South-western Highlands-Mount Kaka
11. All are economic potentials of Afar triangle **except**
- A. geothermal energy C. Irrigation agriculture
 B. exploitation of petroleum & natural gas D. Salt extraction
12. All are the positive impacts of Ethiopia's highly dissected landscape character **except**
- A. Hydroelectric power potential C. Transportation and communication
 B. Settlement expansion D. Agricultural practices
13. Which one of the following is the major threat to the Ethiopian lakes?
- A. Sedimentation C. Afforestation
 B. Water overflow D. Salinization
14. Which one of the following is the highest mountain peak found in the Southwestern highlands of Ethiopia?
- A. Mount Ras Dashen C. Mount Gughe
 B. Mount Tulu Dimtu D. Mount Gara-Muleta
15. Which one of the following is the most irrigated river basins in Ethiopia?
- A. Awash B. Omo
 B. Abay D. Tekeze
16. Which of the following river has unique flow direction?
- A. Tekeze B. Abay C. Baro-Akobo D. Ghibe (Omo).
17. Among the following which Climate Change Response Mechanism is potentially implemented by Ethiopia different from others?
- A. Practice Energy efficiency
 B. Increase the use of renewable energy
 C. Efficient means of transport implementation;
 D. Building flood defenses.
18. One of the following is false about summer solstice
- A. It occur at June 21
 B. North Pole tilted toward the sun
 C. Sub solar point is at 23.5 degree south

Part III: Multiple Choice: Choose the correct answers from the given alternatives (one mark each).

1. The spatial distribution of temperature in Ethiopia is primarily determined by
A. Altitude and latitude C. latitude and cloud cover
B. Distance from the Sea D. distribution of Land and water
2. All are true about Afar triangle **except**
A. It is the widest part of the rift valley C. It is the largest part of the rift valley
B. It is generally milder and marshy D. It is lowest part of the rift valley
3. Wet summer rainfall regions are located in
A. Windward side of mountains C. Leeward side of mountains
B. High altitudes D. All except B
4. During which event does the Northern Hemisphere experience the longest daylight of the year?
A. The summer solstice B. The autumn equinox
B. The winter solstice D. The vernal equinox
5. What atmospheric phenomenon mainly controls the rainfall of Ethiopia?
A. El Nino-Southern Oscillation (ENSO) C. Polar jet stream
B. Gulf Stream D. Intertropical Convergence Zone
6. In Ethiopia, which agro-ecological zone has the largest land coverage?
A. Dega B. Kolla C. Weyna Dega D. Wurch
7. During autumn, which wind is strong and provide rainfall to parts of Ethiopia?
A. Southeasterlies C. Northeast trade wind
B. Equatorial westerlies D. Guinea monsoon
8. Which physiographic sub division is the smallest and has the largest proportion of elevated ground?
A. Southwestern Highlands C. North Central highlands
B. Central highlands D. Southeastern Highlands
9. Which part of the Ethiopian rift valley is characterized by its high altitude and better economic performance is:
A. The Afar Triangle C. The Chew Bahir
B. The Central rift D. The Danakil depression

Part I: True or False: Write "True" if the statement is correct and write "False" if the statement is incorrect statements (one mark each).

- ☒ 1. Ethiopia has the largest proportion of elevated landmass in the African continent and described as the Roof of East Africa
- ☒ 2. The economic importance of the Afar depression includes salt extraction, irrigation agriculture and geothermal energy
- ☒ 3. The lakes region of the Ethiopian Rift valley area has the highest potential for tourism and geothermal energy production
- ☒ 4. Lowlands of Ethiopia are densely settled with sedentary life and permanent settlements as compared to Highlands
- ☒ 5. Awash and Gibe rivers are inland drainage systems that flow within the Rift-valley system of Ethiopia
- ☒ 6. The Inter-tropical Convergence Zone (ITCZ) is a high pressure area where winds Converge

Part II: Matching: Match the items in "Column A" with the correct items in "Column B" (one mark each).

Column: A

- ☒ 1. Formed from Mesozoic limestone rock
- ☒ 2. The lowest point in Ethiopia
- ☒ 3. Afro-alpine summit of Senetti plateau
- ☒ 4. The largest river in volumetric discharge
- ☒ 5. The most exploited river in Ethiopia

Column: B

- A. Kobar Sink
- B. Sof Omar cave
- C. Tulu-Demtu Mountain
- D. Awash River
- E. Abay River
- F. Wabishebele River
- G. Ras-Dejen Mountain